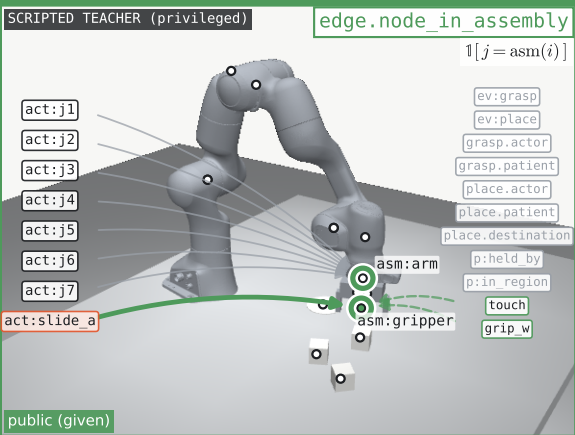
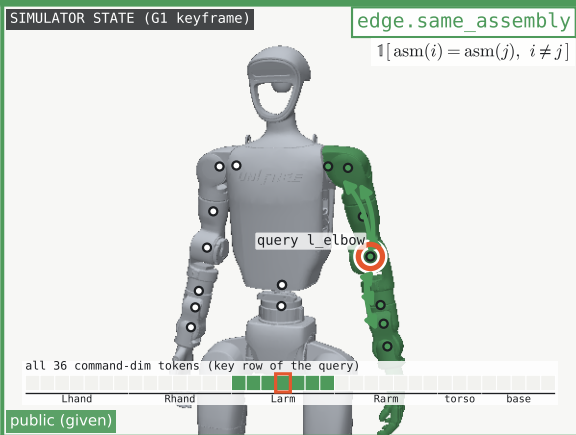


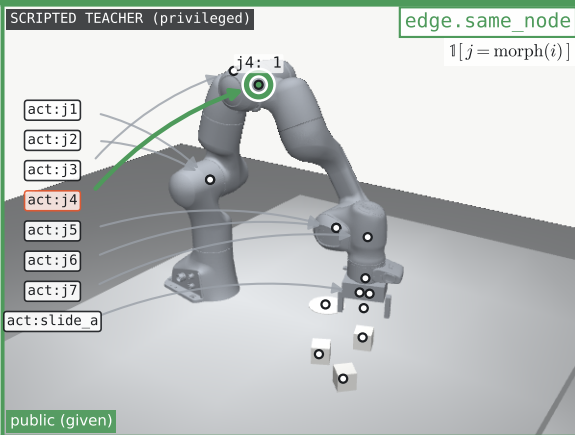
STRUCTURE / EDGES (9 implemented)



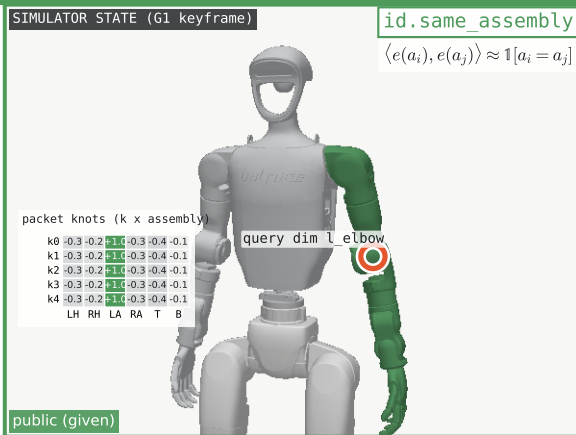
edge.node_in_assembly: a joint / sensor token attends to the assembly (arm, gripper) it belongs to. edge x bias at act>ctx and ctx>ctx (solid: act nodes, dashed: touch / grip sensors; w=1). Source: public morphology (featurizer).



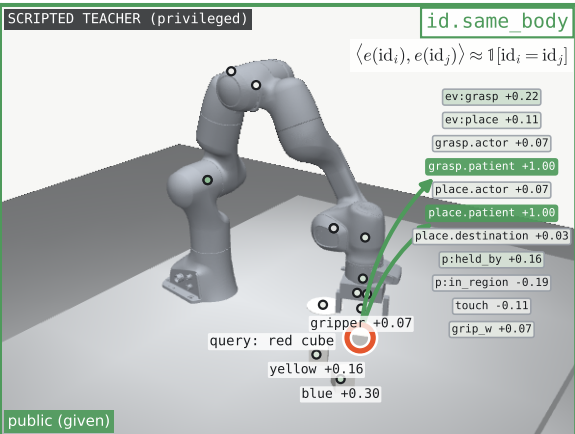
edge.same_assembly: a command dim attends to the other dims of its own assembly (here the left arm). edge x bias at dims>dims (g1-dim-rel-v1 channel same_assembly; w=1). Source: public morphology (bodies.g1_simple).



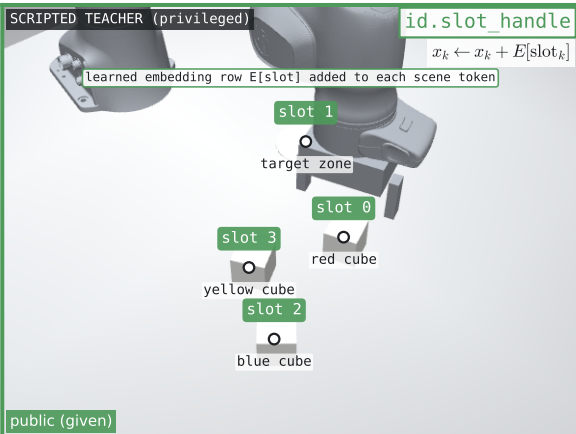
edge.same_node: an action-node query attends to its OWN morphology token (one per actuated joint). edge x bias, w*A[i,j] at act>ctx (arm-rel-v1, w=1 shown). Source: public morphology (featurizer).



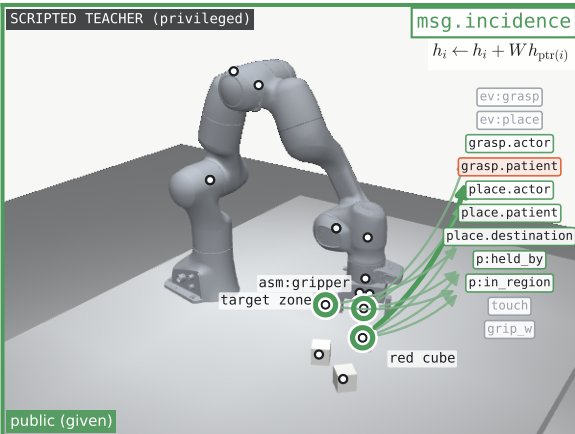
id.same_assembly: soft same-assembly term: a command dim prefers knots of its own assembly (left arm); other assemblies stay readable. same x aug, random 16-d codes for 64 ids (small cross-talk, g=1, dims>knots. Source: public morphology (bodies.g1_simple).



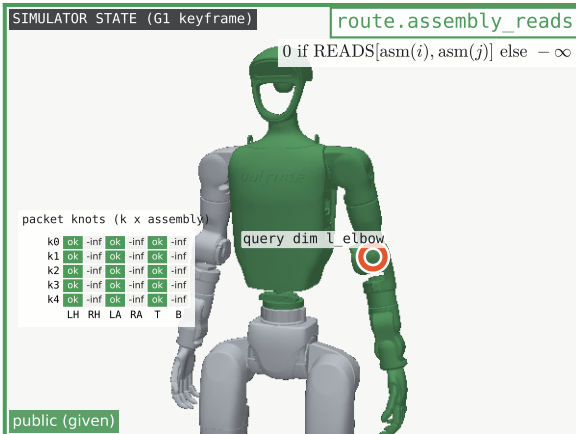
id.same_body: tokens denoting the SAME entity (a tracker slot and the task roles bound to it) attend to each other. same x aug, fixed random 16-d unit codes for 64 ids (1 on equal ids, small cross-talk otherwise), g=1. Source: public ids (featurizer pointers).



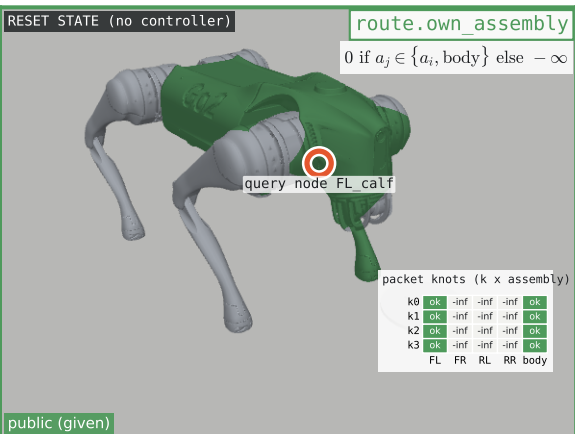
id.slot_handle: each scene token carries its public tracker slot address as a learned embedding, so a slot keeps its identity across steps. embed form (unary, no attention term); E learned. Source: public tracker slot ids.



msg.incidence: role and predicate tokens receive a message from the entity they point to (arrows: entity -> role), message form, not a logit term; W learned; control 'serialized' = same facts as hashed text. Source: public task graph + tracker slots.

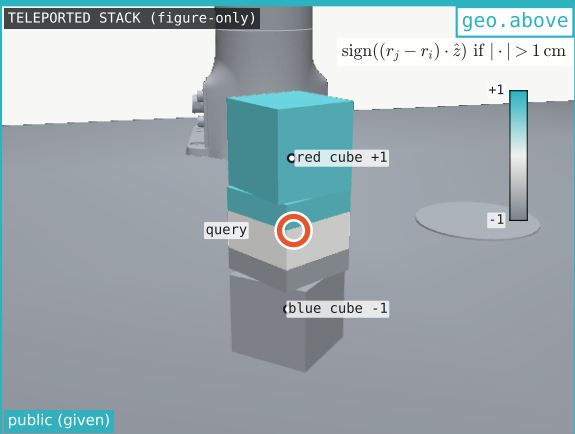


route.assembly_reads: system 0 (Psi0 Realizer): a command dim reads packet knots of its own assembly and its kinematic neighbours only (left arm: arm, hand, torso). same x mask with params.reads = bodies.g1_simple.READS. Source: public morphology (bodies.g1_simple).

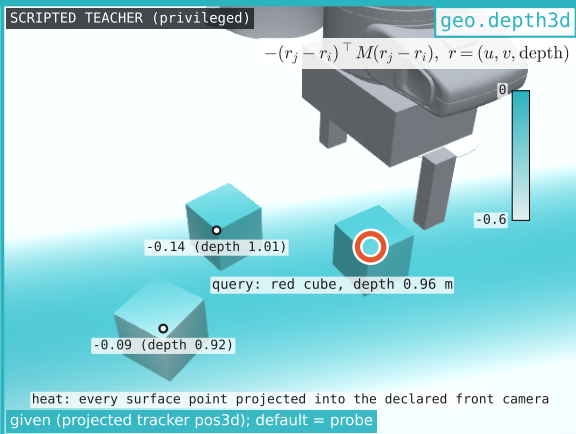


route.own_assembly: system 0 (legged Realizer): a joint node reads only the packet knots of its own limb plus the body assembly. same x mask on assembly_id, params.also_key_field = body (preset legged-s0). Source: public morphology (LeggedMorph).

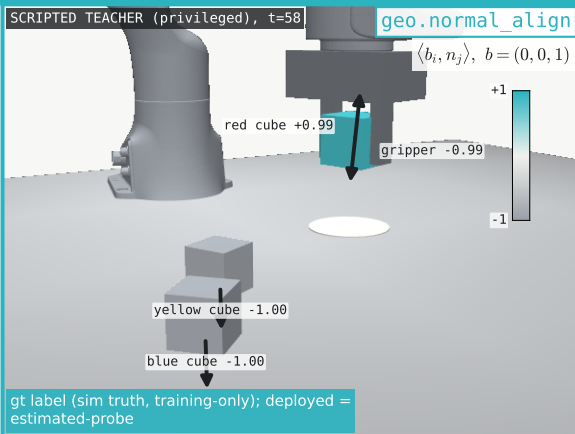
GEOMETRY (5 implemented)



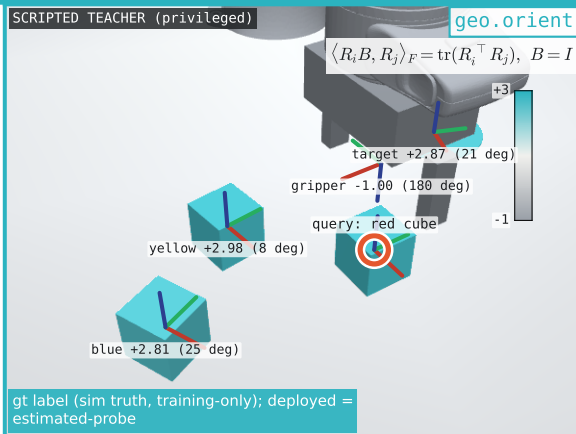
geo.above: above / below along gravity: +1 for tokens (and surface points) higher than the query by > 1 cm, -1 lower, 0 in the dead zone (query: middle cube of a stack). order x bias on pos3d, axis +z, margin 0.01 m (w=1). Source: tracker pos3d (given).



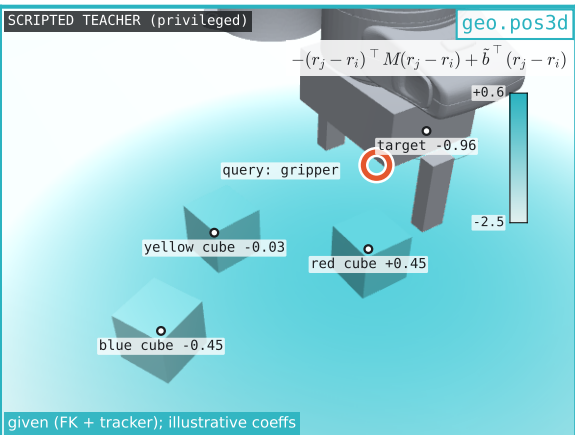
geo.depth3d: PaPE over (u, v, depth) in the declared front camera; weight on depth only, so tokens at the query's depth score high. sqdiff+diff x aug on cam_uvd, a=(0,0,60), b=0 (learned). Source: projected tracker pos3d (given); default source = probe.



geo.normal_align: query direction b vs each token's contact normal (arrows: outward normal, mean over the token's contacts). align x aug, world frame, b=(0,0,1) (learned per head). Source: gt contact_normal (sim truth, training-only); deployed = probe.



geo.orient: relative rotation between token frames: $\text{tr}(R_i^\top R_j) = 1 + 2 \cos(\text{angle})$ (axes: x red, y green, z blue). rel_rot x aug, B = I (learned per head). Source: gt orient label (sim truth, training-only); deployed = probe estimate (object frames are not tracked).



geo.pos3d: PaPE relative-position term: tokens near and below the gripper score high; heat = the term at every surface point. sqdiff+diff x aug, a=30, b=(0,0,-8), W_p=I (learned per head). Source: public FK + tracker pos3d (given).